

```
In [1]: import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
import numpy as np
```

```
In [2]: (x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load
```

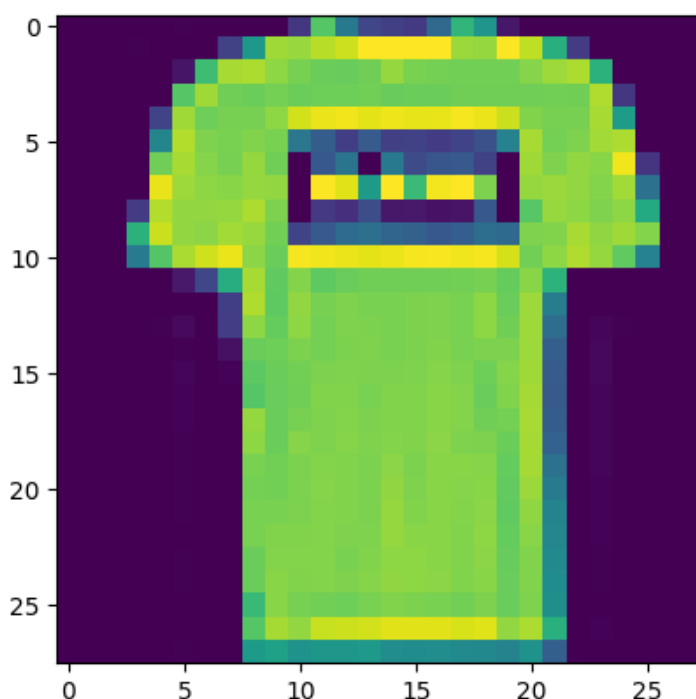
```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz (https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz)
29515/29515 [=====] - 0s 3us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz (https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz)
26421880/26421880 [=====] - 54s 2us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz (https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz)
5148/5148 [=====] - 0s 0s/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz (https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz)
4422102/4422102 [=====] - 8s 2us/step
```

```
In [3]: # There are 10 image classes in this dataset and each class has a mapping
```

```
#0 T-shirt/top #1 Trouser
#2 pullover
#3 Dress
#4 Coat
#5 sandals
#6 shirt
#7 sneaker
#8 bag
#9 ankle boot
```

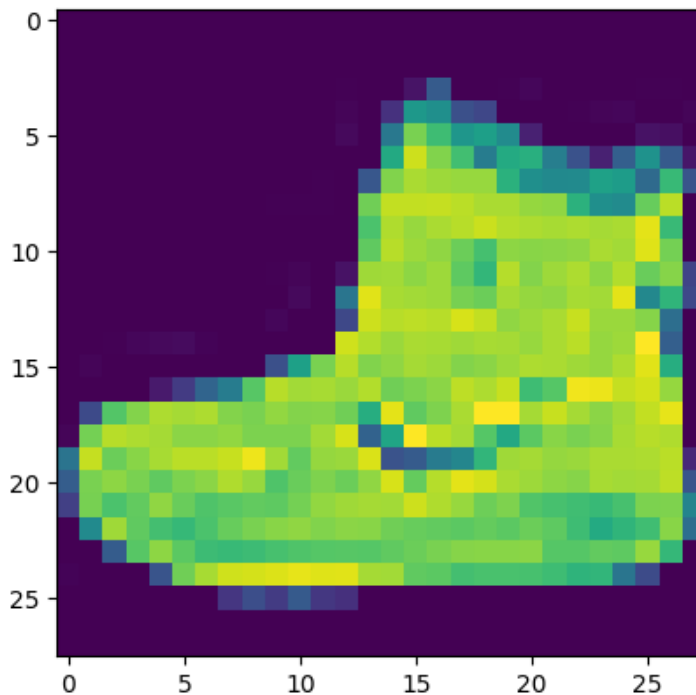
```
In [4]: plt.imshow(x_train[1])
```

```
Out[4]: <matplotlib.image.AxesImage at 0x21909fb7b90>
```



```
In [5]: plt.imshow(x_train[0])
```

```
Out[5]: <matplotlib.image.AxesImage at 0x21909fd2a10>
```



```
In [6]: x_train = x_train.astype('float32') / 255.0  
x_test = x_test.astype('float32') / 255.0  
  
x_train = x_train.reshape(-1, 28, 28, 1)  
x_test = x_test.reshape(-1, 28, 28, 1)
```

```
In [7]: x_train.shape
```

```
Out[7]: (60000, 28, 28, 1)
```

```
In [8]: x_test.shape
```

```
Out[8]: (10000, 28, 28, 1)
```

```
In [9]: y_train.shape
```

```
Out[9]: (60000,)
```

```
In [10]: y_test.shape
```

```
Out[10]: (10000,)
```

```
In [13]: model = keras.Sequential([keras.layers.Conv2D(32, (3,3), activation='relu',  
keras.layers.MaxPooling2D((2,2)),keras.layers.Dropout(0.25),keras.layers.  
keras.layers.MaxPooling2D((2,2)),keras.layers.D  
keras.layers.Flatten(),  
keras.layers.Dense(128, activation='relu'), keras.layers.Dropout(0.25), k
```

In [14]: `model.summary()`

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
dropout (Dropout)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
dropout_1 (Dropout)	(None, 5, 5, 64)	0
conv2d_2 (Conv2D)	(None, 3, 3, 128)	73856
flatten (Flatten)	(None, 1152)	0
dense (Dense)	(None, 128)	147584
dropout_2 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1290

=====
Total params: 241,546
Trainable params: 241,546
Non-trainable params: 0
=====

```
In [18]: model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'],
history = model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test)))

Epoch 1/10
1875/1875 [=====] - 29s 15ms/step - loss: 0.2262 - accuracy: 0.9158 - val_loss: 0.2546 - val_accuracy: 0.9120
Epoch 2/10
1875/1875 [=====] - 32s 17ms/step - loss: 0.2189 - accuracy: 0.9183 - val_loss: 0.2540 - val_accuracy: 0.9093
Epoch 3/10
1875/1875 [=====] - 35s 19ms/step - loss: 0.2108 - accuracy: 0.9196 - val_loss: 0.2769 - val_accuracy: 0.9074
Epoch 4/10
1875/1875 [=====] - 36s 19ms/step - loss: 0.2109 - accuracy: 0.9201 - val_loss: 0.2486 - val_accuracy: 0.9113
Epoch 5/10
1875/1875 [=====] - 40s 21ms/step - loss: 0.2047 - accuracy: 0.9231 - val_loss: 0.2647 - val_accuracy: 0.9104
Epoch 6/10
1875/1875 [=====] - 38s 20ms/step - loss: 0.2027 - accuracy: 0.9228 - val_loss: 0.2584 - val_accuracy: 0.9106
Epoch 7/10
1875/1875 [=====] - 38s 20ms/step - loss: 0.2015 - accuracy: 0.9235 - val_loss: 0.2454 - val_accuracy: 0.9124
Epoch 8/10
1875/1875 [=====] - 38s 20ms/step - loss: 0.1944 - accuracy: 0.9267 - val_loss: 0.2629 - val_accuracy: 0.9111
Epoch 9/10
1875/1875 [=====] - 39s 21ms/step - loss: 0.1916 - accuracy: 0.9265 - val_loss: 0.2631 - val_accuracy: 0.9089
Epoch 10/10
1875/1875 [=====] - 40s 21ms/step - loss: 0.1924 - accuracy: 0.9261 - val_loss: 0.2452 - val_accuracy: 0.9132
```

```
In [23]: test_loss, test_acc = model.evaluate(x_test, y_test)
print('Test accuracy:', test_acc)
```

```
313/313 [=====] - 1s 5ms/step - loss: 0.2452 - accuracy: 0.9132
Test accuracy: 0.9132000207901001
```

```
In [ ]:
```